

Rigorous refraction and reflection laws at interface with a multilayer absorbing medium

Thibault J.-Y. Derrien, Jörg Krüger, and Jörn Bonse
 BAM Bundesanstalt für Materialforschung und -prüfung,
 Unter den Eichen 87, D-12205 Berlin, Germany
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CONTENTS

Introduction	1
I. Experimental conditions of Ref. [1]	1
II. Theory: Model of reflectivity	1
III. Results	2
A. FDTD study of refracted angle as a function of incident angle in absorbing material	2
B. Reflectivity using Snell law of refraction for absorbing materials	2
C. Comparison with experiment	4
D. Taking into account the melted layer index	5
References	5

INTRODUCTION

Aims of this report are:

- to validate an important multilayer reflectivity model by reproducing the experimental results of Sokolowski-Tinten and von der Linde [1],
- to provide a general description of the underlying assumptions, especially on refraction law in absorbing media,
- to address the problem of defocusing in condition of overcritical plasma density.

I. EXPERIMENTAL CONDITIONS OF REF. [1]

In this report, the surface reflectivity is calculated as a function of laser fluence using the laser parameters of Sokolowski-Tinten and von der Linde [1]. The considered processes are the following:

- one-photon ionization, and two-photon ionization : two-photon rate is really high.
- free-carrier absorption
- band-filling effect (absent from free-carrier balance equation, absent from intensity calculation, present in dielectric functions)
- Dielectric constant is not exactly the same as Palik

- Collision frequencies are different for Fresnel and Multilayer equation cases
- Constant band gap

II. THEORY: MODEL OF REFLECTIVITY

Surface reflectivity is defined by $R = |r_{0,N}|^2$ where N is the number of considered optical layers and layer 0 corresponds to the external environment. The surface reflectivity is described by the the Fresnel equations [2, 3]. We take into account the contribution of each layer to the surface reflectivity by using a recurrence formula given by the transmission matrix model [2, 4?]

$$r_{i,k} = \frac{r_{i,i+1} + r_{i+1,k} e^{2i\phi_{i+1}}}{1 + r_{i,i+1} r_{i+1,k} e^{2i\phi_{i+1}}} \quad (1)$$

where

$$\phi_{i+1} = \frac{2\pi h_{i+1}}{\lambda} \sqrt{\varepsilon_{i+1}} \cos \theta_{i+1} \quad (2)$$

is the phase shift induced by the $i + 1$ layer, h_{i+1} is the thickness of the layer $i + 1$, and θ_{i+1} is the angle of incidence at layer $i + 1$. $r_{i,i+1}$ is given by Fresnel formulas [5]

$$r_{i,i+1} = \frac{\sqrt{\varepsilon_i} \cos \theta_i - \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}}{\sqrt{\varepsilon_i} \cos \theta_i + \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}} \quad (3)$$

in the case of perpendicular polarization, and

$$r_{i,i+1} = \frac{\varepsilon_{i+1} \cos \theta_i - \sqrt{\varepsilon_i} \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}}{\varepsilon_{i+1} \cos \theta_i + \sqrt{\varepsilon_i} \sqrt{\varepsilon_{i+1} - \varepsilon_i \sin^2 \theta_i}} \quad (4)$$

in the case of a parallel polarization. ε_i is the full complex dielectric function calculated at the layer i , and θ_i is the angle of incidence at the layer i .

Despite that this point is never mentioned in the papers claiming [1, 4, 6] the use of the multilayer model for reflectivity, a description of the refracted angle is required, especially in the regime of interaction investigated by Sokolowski-Tinten and von der Linde [1]. The well-known law for refraction is given by Snell-Descartes law

$$\theta_{i+1} = \arcsin \left(\sqrt{\frac{\varepsilon_i}{\varepsilon_{i+1}}} \sin \theta_i \right).$$

	Laser wavelength	Pulse duration	Angle of incidence	Polarization
Pump pulse	625 nm	100 fs	30°	S (0°)
Probe pulse	625 nm	100 fs	49°	P (90°)

Table I: Laser parameters taken by Sokolowski-Tinten and von der Linde [1].

	Fresnel reflectivity	Multilayer reflectivity
Collision frequency	0.5 fs	1.1 fs

Table II: Material parameters taken by Sokolowski-Tinten and von der Linde [1].

Snell-Descartes law must be valid in case of complex dielectric functions as stated by Stratton [7] (and experimentally proved by ###). In case of total reflection, a phase shift between equiphase plans and constant amplitude plans appears, leading to heterogeneous waves, then to interferences. Since our multilayer reflectivity model is limited to a homogeneous distribution of fields in the plane normal to the surface, then we must impose a physical limit by which is expressed by the following condition

$$\left| \sqrt{\frac{\epsilon_i}{\epsilon_{i+1}}} \right| \leq 1$$

This work aims to correctly describe the refraction angle as a function of the incident angle and validate a general multilayer model, valid in both transparent and absorbing media.

III. RESULTS

A. FDTD study of refracted angle as a function of incident angle in absorbing material

FDTD simulations have been performed in order to clarify the real refraction angle as a function of the material dielectric functions. The optical index $n + ik$ have been sweep. Figure 1 presents the refracted angle as a function of real optical index n , and for several values of k .

Figure 1 clearly demonstrates that Snell law is not valid in the case of absorbing material, and that a special treatment has to be performed, as stated by Stratton [7].

B. Reflectivity using Snell law of refraction for absorbing materials

In this section, we investigate the different possible approximations in order to find a reasonable description

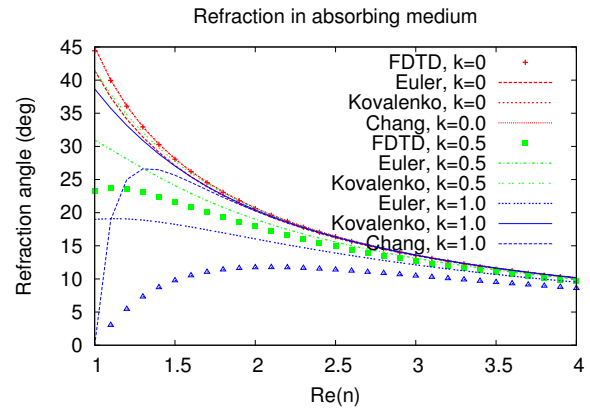
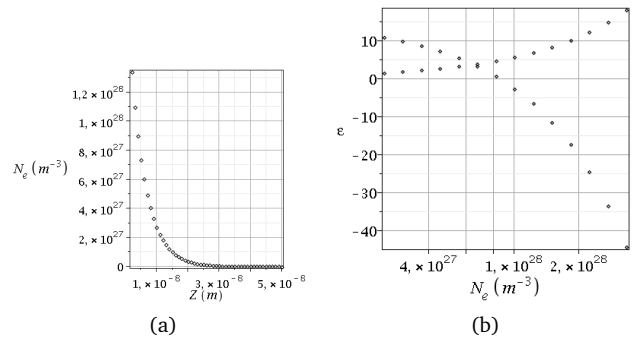


Figure 1: Comparison between FDTD simulation of the refraction angle with Snell law calculated in an absorbing medium.

Figure 2: Profiles considered in the following examples. (a) Free-carrier density. (b) Corresponding dielectric function with $\nu^{-1} = 1.1$ fs.

of the surface reflectivity using the multilayer model, and comparing different description of the refracted angle. First, we prepare a set of distributions to perform the tests.

Figure 2 presents the profile of density and thus of

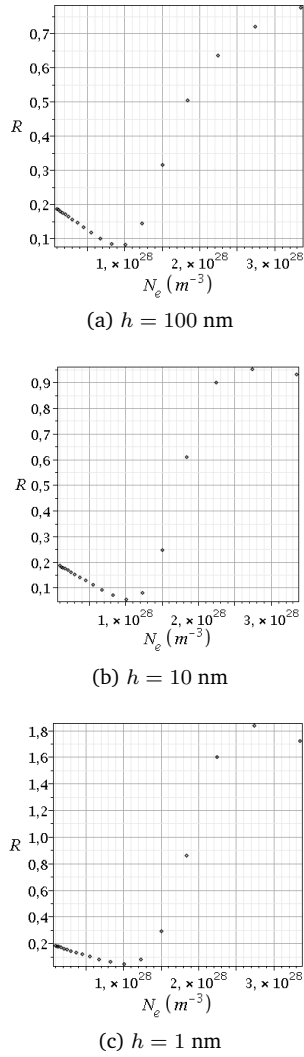


Figure 3: Reflectivity as a function of free-carrier density in Si. Case $\theta_i = \theta_0$.

dielectric function distribution as a function of depth. Free-carrier density is defined by

$$n_e(z) = n_e^0 e^{-\alpha z}$$

where $n_e^0 = 1.35 \times 10^{28} \text{ m}^{-3}$, and $\alpha \simeq 10^7 \text{ m}^{-1}$.

Now, we consider the different models.

a. No refraction approximation Figure 3 presents the reflectivity as a function of free-carrier density using $\theta_0 = \theta_i$ for any layer. As soon as the gradient of density is too strong and that the optical layer is small (~ 10 nm), the reflectivity out-passes physical limits. Then, the description using a real angle of incidence is false. A deeper look at the Eq. (2) shows that even using a real angle, the dielectric function is complex and thus leads to a complex phase, e.g. can lead to an enhancement or damping of the reflectivity in each layer. Indeed, [24] considers that the angle θ_i is always real, but the phase ϕ_i is complex due to the absorbing medium.

b. Refraction in the first layer only Figure 4 presents the reflectivity considering a constant angle of

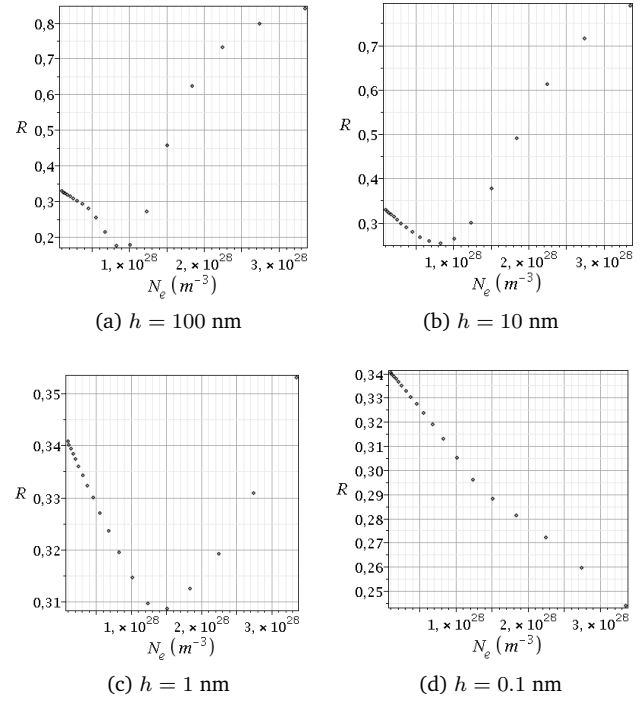


Figure 4: Reflectivity as a function of free-carrier density in Si. Case $\theta_i = \theta_1$.

refraction distribution equal to the one at the surface in the whole sample. The refraction angle at 49° for a density of $3.35 \times 10^{28} \text{ m}^{-3}$ at the surface is 11.5° .

The results show that having a strong gradient lead to a decrease of the reflectivity. The medium is very dense but is also optically very thin.

c. Refraction in each layer; real angles, complex phase This case corresponds to the description of [24], p. 24] and is coded in our simulation code. Reflectivity is a complex quantity, the phase information is also a complex quantity, but the refraction angle is made real before treatment.

d. Refraction in each layer: complex angles, complex phase, complex functions, no limit. Stratton [7] clearly indicates that complex angles are physical and also mention an experimental demonstration.

Figure 5 presents the behavior of the surface reflectivity considering the exponentially decreasing free-carrier density previously defined.

Figure 6 presents the problem in details. This problem originates from a non-physical element present in Eq. (2). Indeed, the term $\cos \theta_i$ with complex valued angle can lead to the case where the following condition is not respected

$$\left| \sqrt{\frac{\epsilon_i}{\epsilon_{i+1}}} \right| \leq 1 \quad (5)$$

thus driving the arcsin function out from its definition set $[-1; 1]$. The angle of incidence is a complex valued quantity since it is used in the Fresnel formulas (Eq. 3,

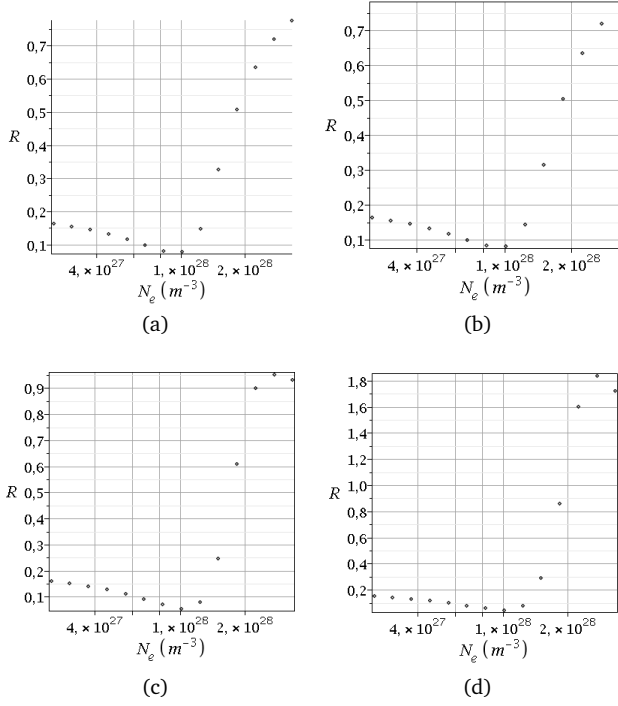


Figure 5: Study of the reflectivity multilayer model. (a) $h_i = 1 \mu\text{m}$. (b) $h_i = 100 \text{ nm}$. (c) $h_i = 10 \text{ nm}$. (d) $h_i = 1 \text{ nm}$.

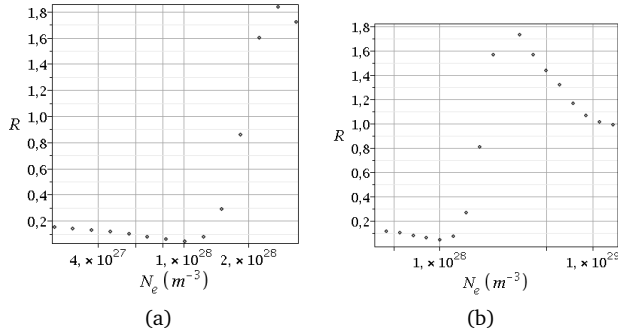


Figure 6: Study of the reflectivity multilayer model. $h_i = 1 \text{ nm}$. (a) $Z^* < 1$. (b) $Z^* > 1$.

4) and lead to a good description only if the refraction leads to lower angles. Thus, we apply Eq. (5) which allows to get the following results.

e. Refraction in each layer: complex angles, phase, and functions, limit enabled. Using a complex angle seems to shift the critical density (defined by $\Re(\varepsilon) = 0$ and $\frac{\partial R}{\partial n_e} = 0$).

C. Comparison with experiment

Figure 8 present the reflectivity as a function of laser fluence. The experimental results have been reported as blue points. The simulation results have been re-

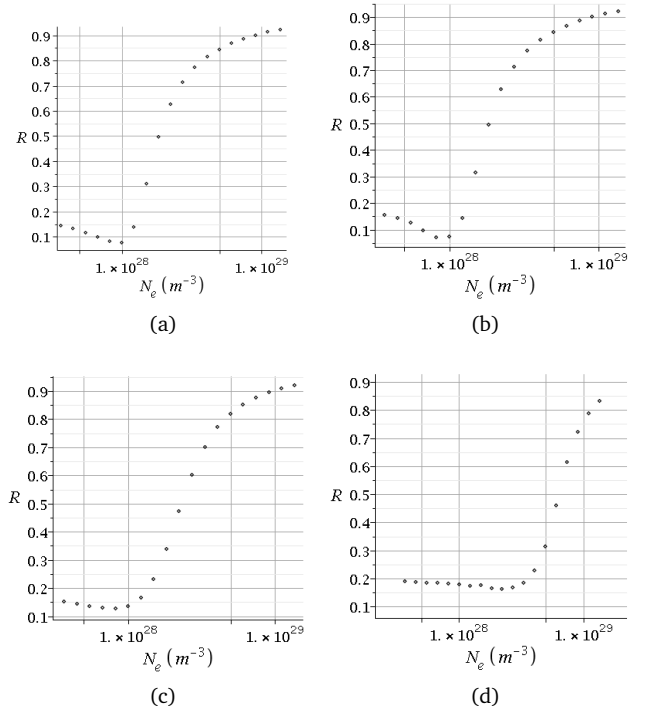


Figure 7: Study of multilayer reflectivity model with a limit given by Eq. (5) on the angle of refraction. $n_e^0 = 2 \times 10^{29} \text{ m}^{-3}$. (a) $h_i = 1 \mu\text{m}$. (b) $h_i = 100 \text{ nm}$. (c) $h_i = 10 \text{ nm}$. (d) $h_i = 1 \text{ nm}$.

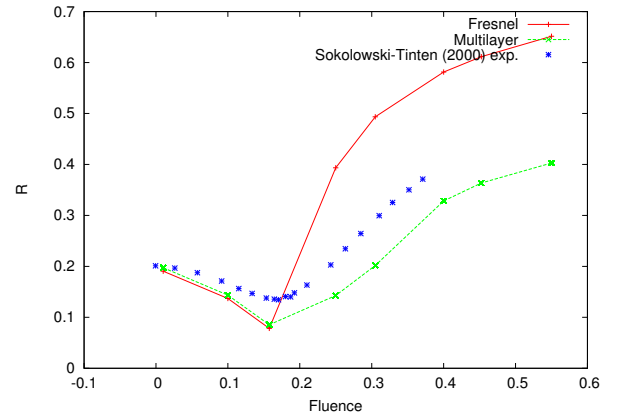


Figure 8: Reflectivity as a function of laser fluence

ported in plain curves. The plain curve corresponds to calculation using Fresnel formula supposing that the refraction medium is optically thick. The dashed curve corresponds to the multilayer model developed in Section II, and assumes that each layer can be optically thin. One observe on this graph that the hypothesis of optically thick medium leads to a mismatch between simulation and experiment. However, multilayer model provide a better match and well reproduces the global behavior of the experiment.

For example, the threshold fluence defined by the

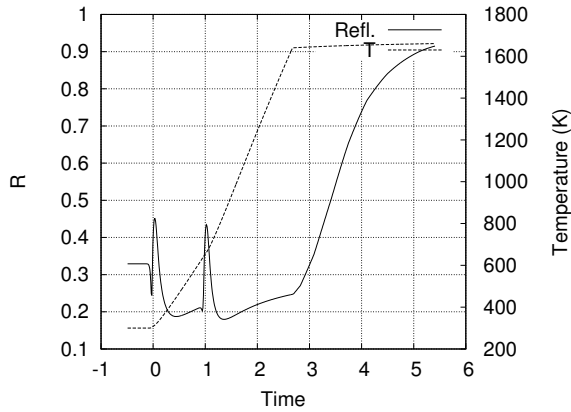


Figure 9: Surface reflectivity evolution and surface lattice temperature after double pulse delay of 100 fs. $F = 0.6 \text{ J/cm}^2$, $\tau = 100 \text{ fs}$, $\lambda = 800 \text{ nm}$.

minimum reflectivity is the same for theory and in the experiment.

However, the minimum reflectivity is slightly different. Indeed, the experimental value is 15%, but the calculated value is 10%. Regarding the knowledge we have on the model analysis we did in the Section II, we can conclude that the gradient of density is stronger in the case of Sokolowski-Tinten calculations. Indeed, a variation of density on a very shallow layer leads to an increase of the minimum reflectivity at the threshold. The gradient of density may be underestimated in the case of the simulation. This can result from the smallest mesh size taken at the surface of the sample, and to the boundary condition.

D. Taking into account the melted layer index

As demonstrated by Sokolowski-Tinten and von der Linde [1], the melting processes are not important in

the time-reflectometry measurements until a certain fluence threshold of nearly 400 mJ/cm^2 , at 625 nm wavelength and 100 fs pulse duration. However, at higher fluences, and especially in double pulse experiments [8] at several ps delays, the liquid layer formation can play a role into the laser absorption. To take this point into account, there exist different formulations. A first model is to take the specular reflectivity change as a function of temperature [9, 10], but has the drawback to not take into account the response of the free-carriers. Then, we propose to base the modeling of surface reflectivity change in melted silicon by calculating the dielectric function of the melted layer following the ratio of melted liquid η in the numerical cell, using the following formula in a first approximation

$$\varepsilon = (\varepsilon_{c-Si} + \Delta\varepsilon_{Drude})(1 - \eta) + \eta\varepsilon_{l-Si}$$

where $\varepsilon_{l-Si} = -14.85 + 47.88i$ [REF?].

Figure 9 presents the evolution of surface reflectivity as a function of time after a double pulse sequence. Laser peak fluence is 0.6 J/cm^2 , pulse duration is $\tau = 100 \text{ fs}$, and laser wavelength is 800 nm. The angle of incidence is normal to the surface. The results demonstrate that the description is not diverging and can be validated on experiments now.

A good question is : what does happen to the free-carrier excited by laser during the solid phase, after transition to liquid state ? The works of Beye et al. [11] would help to answer this question.

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